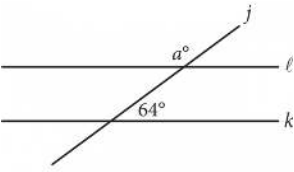


Question ID: 992f4e93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

Answer

- A. 26
- B. 64
- C. 116
- D. 154

Correct Answer: C

Rationale

Choice C is correct. Since lines ℓ and k are parallel, corresponding angles formed by the intersection of line j with lines ℓ and k are congruent. Therefore, the angle with measure a° must be the supplement of the angle with measure 64° . The sum of two supplementary angles is 180° , so $a = 180 - 64 = 116$.

Choice A is incorrect and likely results from thinking the angle with measure a° is the complement of the angle with measure 64° . Choice B is incorrect and likely results from thinking the angle with measure a° is congruent to the angle with measure 64° . Choice D is incorrect and likely results from a conceptual or computational error.